

Priced In, Bit by Bit: Measuring Information Aggregation in Prediction Markets

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Prediction markets are typically evaluated on whether their prices are well calibrated forecasts of eventual outcomes. This paper asks a different question: how informative are prices? Using Shannon entropy and mutual information on a panel of roughly 50,000 closed binary markets on Polymarket, we find that observing prices on average resolves 38.3% of uncertainty surrounding outcomes, and less in markets that are part of a set of mutually exclusive outcomes. Among markets open for over four months, which are more informative than average, prices reduce 42.2% of uncertainty four months from resolution, rising to 85.4% one day from resolution. Finally, following Samuelson’s martingales, we show that while price changes are statistically predictable, any departure from a fair game is economically negligible. These findings are largely consistent with Hayek’s claim that prices aggregate dispersed information.

Keywords: prediction markets, information aggregation, market efficiency, mutual information, entropy

JEL Classification: D83, G14, C13

1. Introduction

Despite their recent surge in popularity, prediction markets are not a novel idea. The earliest recorded example of a ‘prediction’ market goes all the way back to 1503, where people bet on the papal elections ([Rhode and Strumpf, 2013](#)). Five centuries later, traders on Polymarket wagered millions on who would ascend to the Papacy. Since ([Hayek, 1945](#)), economists have recognized that prices serve dual roles: they both allocate resources and aggregate information on the value of those resources. Prediction markets aim to exploit the information aggregating power of prices in an attempt to forecast future events, as argued by a wide range of leading economists ([Arrow et al., 2008](#)).

A price $p \in [0, 1]$ in a prediction market tells us how likely the market believes an outcome Y is. We’ll restrict ourselves to binary market outcomes only, so $Y \in \{0, 1\}$ (either the outcome doesn’t happen, or it does), with $Y = 1$ priced at p and therefore $Y = 0$ priced at $1 - p$. The equilibrium price p equals (or is very close to) the mean beliefs of the traders that the event will occur ([Wolfers and Zitzewitz, 2006](#)). The first modern academic prediction market was the University of Iowa’s Iowa Electronic Markets, introduced during the 1988 US presidential election. Although prediction markets contain many irrational or mistake-prone traders, empirical evidence from the Iowa Electronic Markets suggests that market prices remain highly informative because marginal traders responsible for price formation are substantially more rational than the average participant and arbitrage disciplines persistent pricing errors ([Oliven and Rietz, 2004](#)).

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Prediction markets have a distinctive property over other financial markets: they are one of the very few financial environments in which market efficiency can be evaluated directly because the hidden state of the world is eventually revealed (Thaler and Ziemba, 1988). Consider a stock X . Is the stock correctly priced today? There's no objective answer to that question. The stock's fundamental value is a theoretical object that remains unobservable forever. On the other hand, consider the event $Y = \text{"party } A \text{ wins the election."}$ Y starts out unknown, but eventually, nature will reveal $Y \in \{0, 1\}$ and thus, Y will be observed.

The fact that prediction market prices can be tested against an observed realization of outcomes fundamentally changes what can be tested empirically. In conventional financial markets, the Efficient Market Hypothesis (Fama, 1970) is typically evaluated only indirectly through implications such as return predictability, abnormal returns, or event studies, since the underlying fundamental value is never observed. Prediction markets instead allow direct evaluation of informational efficiency because realized outcomes become available after market resolution.

Existing evaluations of prediction markets have focused primarily on calibration, scoring rules, and forecast accuracy. While these establish whether prices are statistically well calibrated, they do not quantify how much uncertainty market prices eliminate about the eventual outcome. This paper adopts an information-theoretic perspective, measuring the reduction in uncertainty about the eventual outcome attributable to market prices through Shannon entropy and mutual information (Shannon, 1948). Hayek's central claim was that prices aggregate dispersed information. If prices aggregate information, then observing prices should progressively make us less uncertain about the future state of the world as we accumulate new information. Shannon defined information itself as a reduction in uncertainty produced by an observation, making Shannon entropy a natural metric to measure how much information prediction market prices contain.

This paper answers three questions regarding the informativeness of prices. Are they informative at all? If yes, how does the informativeness of prices change over time, and finally, how do prices become informative in the first place? First, we quantify how much uncertainty Polymarket prices resolve on average using mutual information. Second, we trace how this informativeness accumulates over time by measuring the mutual information across various time to resolution bins. Third, we use (Samuelson, 1965)'s martingales to test whether price changes can be predicted, which would negate efficient incorporation of information.

We find that prices resolve roughly 38.3% of outcome uncertainty on average. Negative Risk markets are less informative, resolving 31.4% of outcome uncertainty compared to 45.3% for other markets. Among markets that've been open for more than 4 months, prices already resolve 42.2% of outcome uncertainty 4 months before the market closes, and this figure rises to 85.4% the day before. We also see that while price changes can be predicted with statistical significance (as a result of the large sample size), the predictable part is small. About a fifth of each price move reverses within three hours, which is too little to profit from unless trading costs are very low.

2. Methodology

2.1. Why information theory?

If we'd like to quantitatively study evidence for the Hayekian claim that prices aggregate dispersed information, a field named "Information Theory" is hardly a surprising choice. Hayek's theory is fundamentally about communicating information. If prices can in fact communicate information, it should leave us better informed about the eventual realization of an event (which prediction markets have the unique feature of eventually observing). In simple terms, prices make us better informed, i.e. less uncertain about what will happen. Shannon's theory provides a natural framework because it gives us a way to quantitatively measure information as the *uncertainty reduced* regarding an outcome. This raises the empirical question of how we can quantify uncertainty and its reduction.

(Shannon, 1948) proved that if we have a set of possible events whose probabilities of occurrence are $p_1, \dots, p_n$, with the probabilities being the only thing we know about which event will occur, then the only measure of how much choice is involved in the selection of the event, i.e. how uncertain we are of the outcome, that satisfies a few mathematically desirable properties is of the form

$$H = -K \sum_{i=1}^n p_i \log p_i$$

with $K = 1$ being defined as the *entropy* of the set of probabilities $p_1, \dots, p_n$. Taking $\log_2$ gives entropy measure a unit he called "bits". Entropy is highest when both outcomes are equally likely, since the outcome is most difficult to predict, and falls as one outcome becomes increasingly certain. Thus, it provides an intuitive quantitative measure of uncertainty. Since Polymarket prices are just probabilities (in the binary case, just p_1 and p_2), we can define the entropy of an outcome Y on Polymarket to be $H(Y)$.

Economic logic tells us market prices provide additional information beyond the unconditional probability of an event. After observing the market price, uncertainty about the eventual outcome should decline if prices aggregate information successfully. We denote this as the *conditional entropy* of event Y given the market price using the notation $H(Y|P)$. Mathematically, $H(Y|P)$ is the average entropy of Y for each value of P , weighted by the probability of getting that particular value of P :

$$H(Y|P) = - \sum_p \sum_{y \in \{0,1\}} \mathbb{P}(Y = y, P = p) \log_2 \mathbb{P}(Y = y|P = p)$$

Now that we have a measure for the unconditional uncertainty of Y and the conditional uncertainty upon observing the price, we can finally mathematically define information as uncertainty reduction. Shannon claimed "the rate of actual transmission [of information], R , would be obtained by subtracting from the rate of production (i.e., the entropy of the source) the average rate of conditional entropy." Since prices act as signals in prediction markets, the information transmitted by it is given by

$$I(Y; P) = H(Y) - H(Y|P)$$

Note that what Shannon defined as R , we'll stick to the modern convention and call this quantity *mutual information*, or just information for convenience. Mutual information will be

our main object of interest in this paper. A large value tells us that observing prices dropped uncertainty by a large amount relative to the uncertainty of the unconditional distribution of outcomes, thus, larger values of mutual information are more desirable.

2.2. Information vs calibration

There have been numerous studies that have concluded that besides a few behavioral biases (like the favorite longshot bias), prediction markets are generally accurate and well calibrated, often outperforming professional forecasters and polls in a variety of statistical tests. Studies that show this have been summarized in (Snowberg et al., 2012). Our objective, however, is different. Rather than asking whether prices are *accurate* forecasts, we ask how *informative* they are. Consumers of economic forecasts care not just about the accuracy of the mean prediction, but the uncertainty attached to it as well. The standard proxy for uncertainty is disagreement among forecasters. As (Snowberg et al., 2012) show, that is a poor proxy for actual uncertainty. Using information theory, we provide an alternative way to measure uncertainty in markets.

The difference between information and calibration isn't merely semantic. Consider a market for an election between two candidates A and B , both equally likely to win, so A is priced at 0.5. Observing the price hasn't given us any new information as the unconditional distribution of outcomes is roughly 50% anyways. Suppose a major news story breaks out and pushes A 's price to 0.95. This gives us a lot of new information as we now know that the market thinks A is almost certain to win. Now consider a market that always prices its outcome at the base rate. If 80% of similar markets resolve No, then it prices all these markets at 0.2. This might give us well calibrated outcomes, but observing the price doesn't tell us anything new beyond the base rate, which we already knew. Calibration tells us whether prices match the outcomes that occur, while information tells us how much observing the price reduces our uncertainty of the final outcome. These are complementary, but fundamentally different dimensions of market quality.

3. Data

3.1. Source and Construction

Our dataset combines information from Polymarket's Gamma API, which gives us cross-sectional market data, and the Polymarket CLOB (Central Limit Order Book), which gives us hourly snapshots of the price history across our markets.

To calculate information-theoretic quantities, we need a well defined random variable whose realizations are eventually observed. This restricts our sample to closed markets only. We further restrict our sample to binary markets, so that for an outcome Y , $p = \mathbb{P}(Y = 1)$ and $1 - p = \mathbb{P}(Y = 0)$. Multi-outcome markets would require a joint distribution over several prices at once, which complicates interpretation without adding much to our central question. As these markets are already resolved, there is no trading activity on them and the Gamma API reports a liquidity of zero on almost all of them. Hence, we do not use liquidity in our analysis.

The initial Gamma dataset contained 603,409 binary resolved markets, with data ranging from the start of Polymarket's Gamma data availability till July 18th, 2026. We then sort the data by volume and only select the top 50,000 markets by volume (roughly 71.5th percentile) to be part of our sample. This ensures that our analysis is not skewed by uninformative low volume

markets. Although we started out with 50,000 observations, we had to drop 14 observations with no volume, 3 observations without a specified market close time and a further 12 observations that did not have a price of 0 or 1, i.e. did not cleanly resolve, leaving us with 49,971 markets in our sample. The market IDs of the sample are then matched to the price history data from CLOB to get a panel with historical price data for each market in our sample. This process leaves us with a panel that is indexed by market ID and timestamp (hourly) and has 43.4 million observations corresponding to 49,971 markets.

For each observation, we compute the time remaining until the market closes as the difference between the observation timestamp and the market’s closing time. Section 4 pools every hour of a market’s life, while Section 5 measures information at fixed horizons of d days before close, using balanced cohorts of markets so that the same markets appear at every horizon. In both cases, mutual information is estimated by grouping prices into 20 bins of width 0.05, since entropy requires a discrete distribution of prices.

3.2. Summary Statistics

Table 1: Market Summary Statistics

Variable	Mean	Median	SD	Min	Max
Trading volume (\$ million)	1.126	0.160	10.984	0.052	1531.479
Market duration (days)	38.727	13.385	64.309	0.088	939.485
Resolved Yes (=1)	0.286	N/A	N/A	N/A	N/A
Negative Risk (=1)	0.624	N/A	N/A	N/A	N/A

Table 2: Panel Summary Statistics

Variable	Mean	Median	SD	Min	Max
Price	0.203	0.070	0.270	0	1
Time to resolution (days)	18.77	6.522	31.034	0.007	294.957

4. Question 1: Are prices informative?

The methodology section has given us a quick primer on a quantitative definition of information. Now it is time to apply that and answer the central empirical question of this paper: how informative are these prediction markets?

4.1. Estimation

Mutual information is a property of a joint distribution of outcomes and prices, so before estimating it we must say which distribution we mean. Our panel contains one observation per market per hour, and markets differ enormously in how long they trade: the median market lasts 13 days, while some last more than two years. If every market-hour counted equally, long-lived

markets would dominate the estimate. We instead let every market count equally. Market i is observed for T_i hours, and each of its hours receives weight $1/T_i$, so that each market's weights sum to one. With N markets, the joint distribution of outcomes and binned prices is

$$\hat{\mathbb{P}}(Y = y, P \in b) = \frac{1}{N} \sum_{i=1}^N \frac{1}{T_i} \sum_{t=1}^{T_i} \mathbf{1}[Y_i = y, p_{it} \in b],$$

and $H(Y)$, $H(Y|P)$ and $I(Y;P)$ are all computed from this one distribution. The resulting quantity can be interpreted as the information contained in the price of a typical market, averaged over that market's trading life.

Price is continuous, so we discretize it into 20 bins of width 0.05 and compute $H(Y|P)$ as the entropy of outcomes within each price bin, averaged across bins using the weights above. Binning can only discard information, so in the population $I(Y; \text{bin}(P)) \leq I(Y; P)$ and our estimates are conservative. In finite samples, plug-in entropy estimates are biased downward, which biases mutual information upward. We apply the [Miller \(1955\)](#) correction, which adds $(K - 1)/(2N \ln 2)$ bits to an entropy estimated from N observations of a variable taking K values. Because every hour of a market shares the same outcome, the independent units are markets rather than market-hours, so N is the number of markets. The correction lowers $I(Y; P)$ by about 0.0003 bits in the full sample.

4.2. Results

Table 3 reports the results. A binary outcome has entropy $H(Y) = -(q \log_2 q + (1 - q) \log_2 (1 - q))$, where $q = \mathbb{P}(Y = 1)$, which reaches its maximum of 1 bit at $q = \frac{1}{2}$. In our sample 28.6% of markets resolve Yes, giving a baseline entropy of $H(Y) = 0.8635$ bits, or 86% of the maximum possible uncertainty for a binary event. Therefore, without looking at prices, outcomes are highly unpredictable.

Conditioning on the price lowers the uncertainty to $H(Y|P) = 0.5329$ bits, so prices carry $I(Y; P) = 0.3303$ bits of information about the eventual outcome. This tells us that observing the price of a typical market at some point in its life resolves 38.3% of the uncertainty about how it will resolve. Prices are clearly informative, although a majority of the uncertainty remains until the outcome is known.

Our choice of weights also matters significantly. Markets differ in their lifespan, so if we directly calculate information on our dataset, markets with longer lifespans show up more in the data, and thus our result will be skewed towards such markets. Therefore, our baseline specification weighted markets based on their lifespan, so each market is counted the same in our result. If we skip this and directly report information, we get the hour-weighted metric as markets with longer hours are weighted more. Doing so reveals prices still carry a similar amount of information, 0.31 bits compared to 0.33, but it is a larger share of the uncertainty, 43.7% compared to 38.3%. This difference is largely due to $H(Y)$ being smaller in longer lived markets, which disproportionately resolve No. Only 19.5% of market hours belong to markets that resolve Yes, despite being 28.6% of all markets. This makes our hour-weighted sample dominated by more predictable markets, thereby starting with less uncertainty.

Table 3: Information in Polymarket prices

	Markets	$\mathbb{P}(Y = 1)$	$H(Y)$	$H(Y P)$	$I(Y; P)$	% of $H(Y)$
Baseline (market-weighted)	49,661	0.286	0.8635 (0.0027)	0.5329 (0.0030)	0.3303 (0.0030)	38.3 (0.3)
Hour-weighted	49,661	0.195	0.7108 (0.0064)	0.4002 (0.0055)	0.3103 (0.0055)	43.7 (0.6)

Notes: Entropies and mutual information are in bits. Prices are grouped into 20 bins of width 0.05. In the market-weighted row, each hour of market i receives weight $1/T_i$, where T_i is the number of hours the market is observed, so each market counts once; in the hour-weighted row, every market-hour counts once. $\mathbb{P}(Y = 1)$ is the weighted share of Yes outcomes. $I(Y; P)$ includes the Miller–Madow correction with N equal to the number of markets, so it is about 0.0003 bits smaller than $H(Y) - H(Y|P)$. Bootstrap standard errors in parentheses, from 1,000 draws resampling markets with replacement.

4.3. Negative Risk Markets

Negative risk is a market structure where a combination of positions can reduce risk to zero and potentially create a guaranteed profit by exploiting mispricings. Polymarket has many such markets, usually when dealing with a set of mutually exclusive outcomes. For example, we could have three separate markets for the same award: movie $A/B/C$ wins the Oscar for Best Picture, with each of the three markets being a binary yes/no market for each movie. Assuming the Academy only awards the title to one movie, which it always has done, then one and only one of these 3 markets will resolve to a yes. If you buy a yes contract on all 3 markets, you are guaranteed a payoff of a dollar from the market that resolves yes, while gaining nothing from those that resolved no.

It is important for Polymarket to keep track of markets tied by negative risk due to the presence of arbitrage. If the cost of a yes contract on all 3 markets is less than a dollar, you’ve got a free profit by buying all three yes contracts. Polymarket has a mechanism to convert no shares into yes shares across such markets, allowing arbitrage to push yes prices towards summing up to 1. This presents a problem for our information aggregation story as now prices can change due to arbitrage, even without the introduction of new information into the market.

Table 4 splits the sample into negative-risk and other markets. The two groups differ sharply in baseline uncertainty. Only 23.6% of negative-risk markets resolve Yes, compared with 36.9% of other markets, so negative-risk outcomes are more predictable to begin with ($H(Y) = 0.788$ against 0.950 bits). This isn’t surprising, as a set of mutually exclusive markets generally has a few genuine contenders and many longshots that are priced near zero and resolve No. As in our Oscars example, even though very few movies are considered serious contenders for Best Picture, markets for many more movies than that may exist on Polymarket.

Prices in negative-risk markets are also less informative. They carry 0.25 bits of information about the outcome, compared with 0.43 bits in other markets, and resolve 31.4% of the uncertainty compared with 45.3%. This gap disappears when every market-hour counts equally, which gives 43.2% and 43.8%. Hour weighting emphasizes long-lived markets, so the difference comes from short-lived markets: among markets that trade for a long time, prices resolve about the same share of uncertainty under both structures, while short-lived negative-risk markets are markedly less informative than other short-lived markets. Whether this reflects arbitrage-driven

Table 4: Information in prices by market structure

	Markets	$\mathbb{P}(Y = 1)$	$I(Y; P)$	% of $H(Y)$ resolved	
				Baseline	Hour-weighted
All markets	49,661	0.286	0.3303 (0.0030)	38.3 (0.3)	43.7 (0.6)
Negative risk	30,999	0.236	0.2473 (0.0033)	31.4 (0.4)	43.2 (0.8)
No negative risk	18,662	0.369	0.4305 (0.0050)	45.3 (0.5)	43.8 (1.0)
Difference			-0.1832 (0.0059)	-13.9 (0.6)	-0.7 (1.3)

Notes: The two groups split the full sample and together make up the “All markets” row. Markets, $\mathbb{P}(Y = 1)$ and $I(Y; P)$ (in bits, Miller–Madow corrected) refer to the baseline, market-weighted specification. The last two columns report the share of $H(Y)$ resolved by prices under the two specifications of Table 3. “Difference” is negative-risk minus other markets. Bootstrap standard errors in parentheses, from 1,000 draws resampling markets with replacement; the standard error of the difference uses the same draws for both groups.

price movements or the kinds of events that short-lived negative-risk markets cover is a question we leave for future work.

5. Question 2: The dynamics of informativeness

The previous section established that market prices eliminate over a third of the uncertainty surrounding future outcomes on average. However, an average alone tells us little about when this information enters prices. It could come months in advance, or when an event happens that makes the outcome obvious. Prediction markets telling us that an event will happen after the general consensus has ruled it as extremely likely is not very informative. An informative market must certainly have the ability to be informative long before the event is considered “settled”.

5.1. Measuring information at a given horizon

We measure how much information the price contains d days before a market closes, denoted by $I(Y; P_d)$ for $d = 1, 2, \dots$ days. For each market and day d , we take the last hourly price recorded at least d days before the market is closed, so each market contributes one observation at each day. Estimation follows the same process as in Section 4, using 20 price bins and Miller-Madow correction.

Before we plot information against time, we must be mindful of a potential selection bias: markets have different lifespans. All closed markets must, by definition of being closed, have existed an hour before resolution, while not all markets need to be 4+ months old. Directly plotting would give us a different sample size in each time to resolution bin. We therefore work with balanced cohorts: the cohort for a lifetime of D days contains the markets that traded for at least D days, and the same markets appear at every horizon from D days to one day. We use four cohorts, with D equal to one week, one month, two months and four months. We also require

every market to have traded for at least 48 hours before the cohort's earliest horizon, so that no market enters the curve on its opening day, when its price is still largely the initial quote and is largely uninformative.

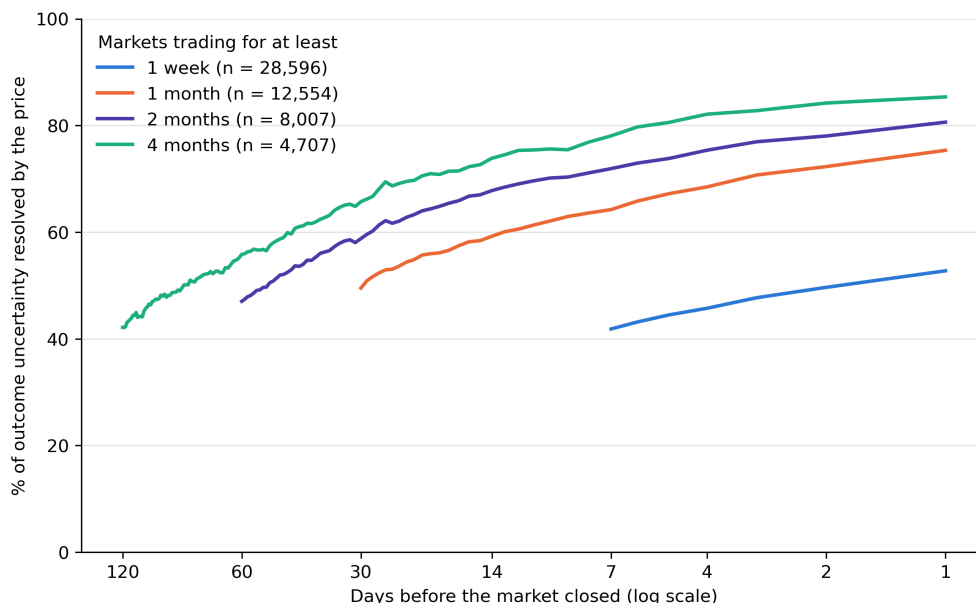


Figure 1: Share of outcome uncertainty resolved by the price, by days before the market closed.

Figure 1 plots the share of outcome uncertainty resolved by the price against the number of days before close, and Table 5 reports the four-month cohort at selected horizons. Prices in long-lived markets are informative early. Four months before closing, prices in the four-month cohort already resolve 42.2% of the uncertainty about the outcome. This share rises steadily to 65.7% one month before close, 78.1% one week before, and 85.4% one day before. Information aggregation is a continuous process that begins long before outcomes become obvious.

At every horizon, prices in longer lived markets resolve more of the uncertainty. A potential explanation is the kind of event each cohort covers. Shorter markets may tend to include more volatile events like close sports games, where the outcome could drastically change a few hours before resolution. On the other hand, longer lived markets deal with questions that prices identify as unlikely well in advance, as only 14.5% of the markets in the 4 month cohort resolve Yes.

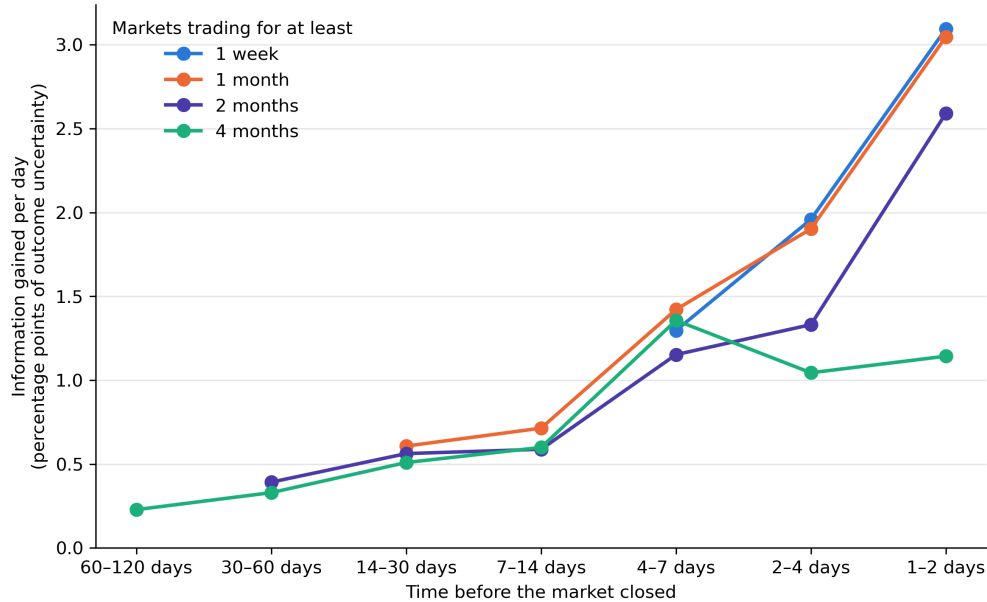
5.2. Rate of information aggregation

To measure changes in information aggregation, we can plot the average information gain per day in each time period against the time periods directly. Simply using the daily change in informativeness is largely noise and is not very informative, thus we bin days into different time periods.

Table 5: Information in prices by horizon, four-month cohort

Days before close	$H(Y P_d)$	$I(Y;P_d)$		% of $H(Y)$		Gain per day	
		Est.	S.E.	Est.	S.E.	Est.	S.E.
120	0.3428	0.2519	0.0118	42.2	1.6	–	–
60	0.2610	0.3336	0.0129	55.8	1.7	0.23	0.02
30	0.2019	0.3926	0.0128	65.7	1.5	0.33	0.04
14	0.1533	0.4413	0.0134	73.9	1.5	0.51	0.07
7	0.1283	0.4660	0.0134	78.1	1.4	0.60	0.13
4	0.1038	0.4899	0.0139	82.1	1.3	1.36	0.26
2	0.0914	0.5024	0.0139	84.2	1.3	1.04	0.31
1	0.0849	0.5096	0.0139	85.4	1.3	1.14	0.59

Notes: Markets that traded for at least four months ($n = 4,707$). $H(Y) = 0.597$ bits at every horizon (14.5% of these markets resolve Yes). $I(Y;P_d)$ includes the Miller-Madow correction. “Gain per day” is the change in the share of $H(Y)$ resolved since the previous row, in percentage points, divided by the number of days between the two horizons. S.E. columns report bootstrap standard errors from 1,000 draws resampling markets in the cohort with replacement.

**Figure 2:** Average information gained per day, in percentage points of outcome uncertainty, in each stretch of time before the market closed.

Prices clearly learn faster as resolution approaches. In the four-month cohort, prices resolve only 0.23 percentage points of uncertainty per day between four and two months before close. Between seven and four days before close it reaches 1.36, roughly six times the rate four months out. Within the final days, however, the estimates become imprecise, and the gains over the last four days, 1.04 (0.31) and 1.14 (0.59), are statistically indistinguishable from each other and from the rate between seven and four days, so we cannot say whether learning levels off or keeps accelerating in the last days before close. The other cohorts show the same rising pattern.

These results show that information aggregation in prediction markets is on average a gradual process, not confined to the moments right before resolution. Markets do contain substantial amounts of information many months before resolution. The average informativeness reported in the previous section is therefore not driven solely by prices immediately before market resolution but reflects information that has already accumulated throughout the market’s lifetime. The results also suggest that the price’s stock of information increases at an increasing rate, although this is hard to definitively establish using just discrete approximations of change.

6. Question 3: How do prices incorporate information?

So far, we’ve used mutual information to show that prices contain substantive information and how that information content increases over time. However, a natural question remains: how does information enter these prices? There are numerous plausible mechanisms. Perhaps news is immediately efficiently incorporated into prices by rational traders. Maybe information is gradually diffused into the market over time. It is also possible that traders may initially overreact to new information, leading to the price overshooting and then returning to an equilibrium. Across all three mechanisms, we could end up with very different price dynamics but similar information trajectories.

One way we can distinguish between these cases is by looking at serial correlation in price changes. Following (Fama, 1970), if prices immediately react to and incorporate new information, then you wouldn’t expect any persistent pattern in price changes. Changes in price would show zero serial covariance (and thus no autocorrelation). On the other hand, gradual diffusion of positive news would lead to a persistent upward pattern in price changes, and overreaction would lead to a correction that shows persistent downward price changes, and vice versa for negative news.

For lags $k \in \{1, \dots, 24\}$, we compute the correlation between the current price change Δp_t and the k -hour lagged price change Δp_{t-k} per market, plotted in Figure 3. Each distribution has a single mode slightly below zero, with a small secondary spike at roughly -0.75 , -0.5 , and -0.25 respectively. These are exactly the coefficients an AR(3) regression gives us when a price fluctuates randomly around a fixed level (Appendix A), suggesting these β s are from markets that are largely settled and have no real information incorporation. Indeed, markets that produce these β s have a median price of 2 cents and spend 97% of their trading hours priced below 5 or above 95 cents (compared with 22% for other markets). Their prices bounce around a nearly fixed value, which is what bid-ask bounce produces (Roll, 1984), rather than slow or excessive reaction to news.

It could be argued that our autocorrelation plot is an artifact of aggregation, where we genuinely have different types of markets that incorporate information differently, yet aggregating

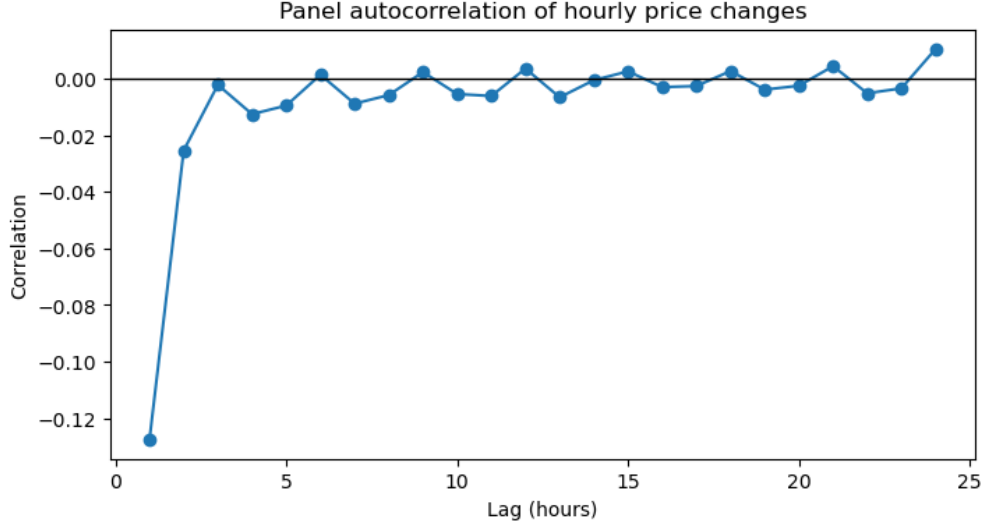


Figure 3: Panel autocorrelation of hourly price changes.

the results smoothens differences and presents it as looking like an informationally efficient market. To test this, we plot the coefficients β_1, β_2 and β_3 obtained from regressing Δp_t on $\Delta p_{t-1}, \Delta p_{t-2}$ and Δp_{t-3} for each market and plot the distribution of β s obtained. We see no signs of multimodal distributions here. This tells us that autocorrelation patterns observed are pervasive and consistent across markets, rather than an artifact of pooling heterogeneous markets. Combined with the very small autocorrelation observed beyond one hour, this points towards markets efficiently incorporating information.

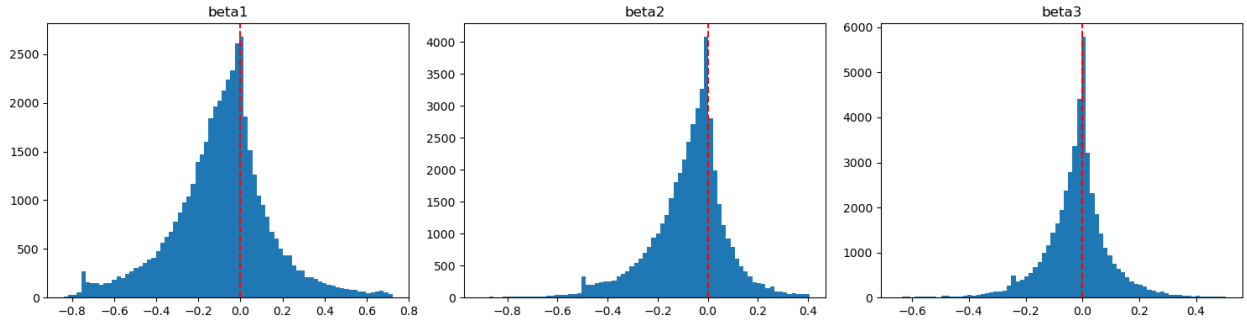


Figure 4: Distribution of autocorrelation coefficients.

6.1. Polymarket prices as a fair game

So far, we've tested whether a market's own price changes can predict future price changes, i.e. Fama's weak form efficiency. A broader result is (Samuelson, 1965), who showed that if prices are formed as the conditional expectation of an outcome given *all available information*, $p_t = \mathbb{E}[Y|I_t]$, then price changes behave like a fair game with respect to the information set, i.e. $\mathbb{E}[\Delta p_t | I_t] = 0$: given everything we know, the expected future price equals today's price. Prices can only change if new information arrives, and by definition that cannot be predicted. This asserts that no piece of information can predict future price changes. On Polymarket, a price change of one cent

reflects a one percentage point change in the market’s assessment of an event’s probability, and is also the profit per share of someone who held their share through that change. Therefore, predicting Δp_t is the same as predicting a trader’s profit, and Samuelson’s fair game condition tells us that the expected profit of any strategy built on known information at time t is zero.

Unfortunately, we do not have access to every single possible datapoint players in the market may use to make their decisions. Our best option is to use a wider set of variables that are known when the price change begins: the previous price, the time remaining until the market’s scheduled end date, recent volatility, and the three most recent price changes. We estimate

$$\begin{aligned} \Delta p_t = & \beta_0 + \beta_1 p_{t-1} + \beta_2 \log(1 + \max(\tau_t, 0)) + \beta_3 \text{Overdue}_t \\ & + \beta_4 \sigma_{t-1} + \sum_{k=1}^3 \rho_k \Delta p_{t-k} + \varepsilon_t, \end{aligned} \quad (1)$$

where Δp_t is the hourly price change in cents, τ_t is the number of hours until the market’s scheduled end date, and σ_{t-1} is the standard deviation of hourly price changes over the previous 24 hours. The price enters with a lag because p_t already contains Δp_t , which would mechanically produce a positive coefficient. Time is measured till the *scheduled* end date because the actual closing time depends on when the event resolves (which needn’t be the scheduled time), something traders don’t know in advance. Markets may trade past their scheduled end dates while the outcome is known, so we use the dummy *Overdue*_{*t*} to capture any differences during these hours. Under the martingale hypothesis, every coefficient in (1) is zero.

Table 6: Predictability of hourly price changes

	Regressor	Coefficient	Std. error
β_1	p_{t-1}	−0.0464	0.0001
β_2	$\log(1 + \tau_t)$	−0.0033	0.0002
β_3	<i>Overdue</i> _{<i>t</i>}	0.0177	0.0048
β_4	σ_{t-1} (cents)	−0.0163	0.0006
ρ_1	Δp_{t-1}	−0.1429	0.0021
ρ_2	Δp_{t-2}	−0.0505	0.0013
ρ_3	Δp_{t-3}	−0.0146	0.0013
β_0	Intercept	0.0374	0.0018

Notes: Pooled OLS estimates of equation (1); the dependent variable is the hourly price change in cents, so each coefficient is the predicted change in a share’s value, in cents per hour. 42,353,513 observations from 48,974 markets; markets without a scheduled end date are excluded. $R^2 = 0.021$; joint Wald $F = 977.161$. Standard errors clustered by market. All coefficients are significant at the 1% level.

42 million observations effectively makes any result statistically significant, and the joint test rejects the hypothesis that all of them are zero. What matters here is economic significance, whether the predictable part is large enough to matter. Since our dependent variable is in cents per share, each coefficient can be interpreted as the change in expected profits.

Almost all the predictability comes from short run reversal. A one cent rise in prices is associated with an expected fall of 0.14 cents in the next hour (ρ_1) and 0.21 cents over the next three hours ($\rho_1 + \rho_2 + \rho_3$), so around a fifth of any move is undone within three hours. To profit from this, a trader must trade against a move as soon as it happens. This is only profitable for a move of m cents if the bid ask spread is below $0.21m$ cents. We do not observe spreads in our data as Polymarket’s CLOB only provides it for live markets, so we cannot rule this out for every period. Since March 2026, Polymarket has themselves charged trading fees on most markets, making exploiting known information profitably even harder. The negative ρ coefficients could also simply be bid–ask bounce instead of genuine overreaction, in which case there really isn’t anything exploitable to begin with.

The remaining coefficients describe slow drifts, all well under a tenth of a cent per hour. A market priced at 90 cents is predicted to rise about 0.04 cents per hour less than one priced at 10 cents ($0.8\beta_1$). Doubling the time remaining until the scheduled end date lowers the predicted hourly change by about 0.002 cents ($\beta_2 \ln 2$). Overdue markets drift up by about 0.02 cents per hour (β_3), and a market whose price has recently been moving two cents an hour is predicted to rise about 0.03 cents per hour less than a still one ($2\beta_4$). Over a full day even the largest of these drifts adds up to under a cent, while holding a share for a day exposes the trader to the full uncertainty of the outcome.

Taken together, the regressors explain 2.1% of the variation in hourly price changes, leaving 97.9% unpredictable from this information set. A low R^2 alone does not show that predictability is unimportant (Campbell and Thompson, 2008), which is why we focus on the magnitudes above. But these magnitudes are small enough that they leave no exploitable strategies based solely on publicly available current information, consistent with Samuelson’s fair game condition.

7. Conclusion

Prediction markets have found their place in mainstream discourse, no longer being a fringe interest among a few economists. Their odds are now reported next to polls on major news networks, with people bidding real money on everything from interest rates to the weekend football game. Advocates of these markets argue that governments and firms should act on the predictions of these markets, as they believe the market “knows” something. This paper sets out to measure how much the market knows, using Shannon’s mutual information on roughly 50,000 resolved binary markets on Polymarket.

This paper asked three questions about prediction market prices: whether they are informative, how their informativeness evolves as resolution approaches, and whether information is incorporated efficiently. We find that observing the price of a typical market resolves 38.3% of the uncertainty about how it will resolve. Prices in negative risk markets are less informative, resolving 31.4% of uncertainty against 45.3% in other markets, a gap that comes from short-lived markets. In markets open for at least four months, prices already resolve 42.2% of the uncertainty four months before close, rising steadily to 85.4% the day before, with prices learning faster as resolution approaches. Information enters prices gradually throughout a market’s life, not only once the outcome has become obvious.

Information enters prices in a manner that is close, but statistically distinguishable from efficient incorporation, with any predictable part in hourly price changes being small. About

a fifth of a price move reverses within three hours and trading against it only pays off if the bid-ask spread is below a fifth of the price move. The other public information we condition on combined predicts drifts of under a tenth of a cent per hour. Thus, we can conclude that not only do prices on Polymarket aggregate dispersed information, what is publicly known is almost fully priced in.

There are a few paths that we can still explore, left for future work. The most obvious is to split markets by topic, as politics, sports, crypto etc. are inherently different kinds of markets and would likely have different informativeness patterns. Aligning price paths with the public release of information, like press conferences, would allow us to measure how quickly prices absorb the new information. Finally, spread snapshots or trade level data would let us measure trading costs directly to better assess whether we can exploit publicly available information profitably.

In conclusion, Polymarket does what its advocates claim to an extent. Its prices aggregate what scattered traders know, steadily absorb new information over a market's lifespan, and leaves little that public information alone can predict. Yet they're not perfect, leaving more than half of the uncertainty unresolved months before resolution, and only resolve about 85% of the uncertainty a day before resolution. Thus a Polymarket price can hardly be considered a verdict, but it sure isn't noise either. It can be best interpreted as a snapshot of what the public knows at a given moment, with the wisdom of the crowd growing more reliable as the outcome draws closer. The market does in fact "know" something, but how much it knows depends on when you ask.

Appendix

A. AR(3) coefficients when prices fluctuate around a constant

Suppose a market is largely settled already, so its price is fixed at some level c , plus a random noise term. This gives us

$$p_t = c + e_t \quad \mathbb{E}[e_t] = 0, \quad \text{Var}(e_t) = \sigma^2$$

Let x_t be defined as $\Delta p_t = e_t - e_{t-1}$, the difference of noise terms and let $\gamma_k = \text{Cov}(x_t, x_{t-k})$. Computing the values of γ gives us

$$\begin{aligned} \gamma_0 &= \text{Var}(e_t) + \text{Var}(e_{t-1}) = 2\sigma^2 \\ \gamma_1 &= -\text{Var}(e_{t-1}) = -\sigma^2 \\ \gamma_k &= 0 \quad \text{for } k \geq 2 \end{aligned}$$

Now consider the regression $x_t = \beta_0 + \beta_1 x_{t-1} + \beta_2 x_{t-2} + \beta_3 x_{t-3} + u_t$. We know that in large samples, OLS gives $\mathbb{E}[u] = 0$ and $\text{Cov}(x_{t-k}, u_t) = 0$. Taking expectations on both sides, $\mathbb{E}[x_t] = \mathbb{E}[e_t] - \mathbb{E}[e_{t-1}] = 0$, which implies $\alpha = 0$, so the intercept drops out. Rearrange to isolate the residual and take covariance of both sides with x_{t-k} . This gives us

$$0 = \text{Cov}(x_t, x_{t-k}) - \beta_1 \text{Cov}(x_{t-1}, x_{t-k}) - \beta_2 \text{Cov}(x_{t-2}, x_{t-k}) - \beta_3 \text{Cov}(x_{t-3}, x_{t-k}).$$

In terms of γ , we get

$$\gamma_k = \beta_1 \gamma_{k-1} + \beta_2 \gamma_{k-2} + \beta_3 \gamma_{k-3}.$$

Since $\text{Cov}(x_t, x_{t-k}) = \text{Cov}(x_{t-k}, x_t)$, $\gamma_k = \gamma_{-k}$. Taking $k = 1, 2, 3$ and dividing out the σ^2 , we get

$$\begin{aligned} -1 &= 2\beta_1 - \beta_2 0 & &= -\beta_1 + 2\beta_2 - \beta_3 0 = -\beta_2 + 2\beta_3 \end{aligned}$$

and solving this system gives us

$$\beta_1 = -\frac{3}{4}, \quad \beta_2 = -\frac{1}{2}, \quad \beta_3 = -\frac{1}{4}$$

which exactly matches the points where the β s pile up in Figure 4. Therefore, these spikes do not represent inefficient markets, and instead represent markets which are largely settled and don't have much price movements besides noise.

References

- Kenneth J. Arrow, Robert Forsythe, Michael Gorham, Robert Hahn, Robin Hanson, John O. Ledyard, Saul Levmore, Robert Litan, Paul Milgrom, Forrest D. Nelson, George R. Neumann, Marco Ottaviani, Thomas C. Schelling, Robert J. Shiller, Vernon L. Smith, Erik Snowberg, Cass R. Sunstein, Paul C. Tetlock, Philip E. Tetlock, Hal R. Varian, Justin Wolfers, and Eric Zitzewitz. [The Promise of Prediction Markets](#). *Science*, 320(5878):877–878, 2008.
- John Y. Campbell and Samuel B. Thompson. Predicting excess stock returns out of sample: Can anything beat the historical average? *Review of Financial Studies*, 21(4):1509–1531, 2008.
- Eugene F. Fama. [Efficient capital markets: A review of theory and empirical work](#). *The Journal of Finance*, 25(2):383–417, 1970.
- Friedrich A. Hayek. [The Use of Knowledge in Society](#). *The American Economic Review*, 35(4):519–530, 1945.
- George A. Miller. Note on the bias of information estimates. In Henry Quastler, editor, *Information Theory in Psychology: Problems and Methods*, pages 95–100. Free Press, Glencoe, IL, 1955.
- Kenneth Oliven and Thomas A. Rietz. [Suckers Are Born but Markets Are Made: Individual Rationality, Arbitrage, and Market Efficiency on an Electronic Futures Market](#). *Management Science*, 50(3):336–351, 2004.
- Paul W. Rhode and K Coleman Strumpf. [The long history of political betting markets: An international perspective](#). In Leighton Vaughan Williams and Donald S. Siegel, editors, *The Oxford Handbook of the Economics of Gambling*, pages 560–583. Oxford University Press, 2013.
- Richard Roll. [A simple implicit measure of the effective bid-ask spread in an efficient market](#). *The Journal of Finance*, 39(4):1127–1139, 1984.
- Paul A. Samuelson. [Proof That Properly Anticipated Prices Fluctuate Randomly](#). *Industrial Management Review*, 6(2):41–49, 1965.
- Claude E. Shannon. [A mathematical theory of communication](#). *The Bell System Technical Journal*, 27(3):379–423, 1948.
- Erik Snowberg, Justin Wolfers, and Eric Zitzewitz. [Prediction Markets for Economic Forecasting](#). Discussion Paper 6720, Institute for the Study of Labor (IZA), 2012.

Richard H. Thaler and William T. Ziemba. [Anomalies: Parimutuel betting markets: Racetracks and lotteries](#). *Journal of Economic Perspectives*, 2(2):161–174, 1988.

Justin Wolfers and Eric Zitzewitz. [Interpreting prediction market prices as probabilities](#). Working Paper 06-11, Rodney L. White Center for Financial Research, The Wharton School, University of Pennsylvania, 2006.